Candidate Seat No:_________________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Science in Information Technology (M.Sc. IT)
Semester-I University Examination November-December 2018
MS 112 Operating System Concepts

Date: 04.12.18, Tuesday   Time: 10.00 a.m. to 01.00 p.m.   Maximum Marks: 70

Instructions:
1. **Attempt both sections in separate answer books.**
2. **The figure to the right indicates marks.**
3. **Make a suitable assumption when necessary.**
4. **Simple Calculator is permitted.**

SECTION-1

Q.1 **Give an appropriate answer from the following Multi Choice Questions(MCQs).** **[11]**

1. In the _______ Real Time System, time is fixed and we can't change any moments of the time of processing.
(a) Soft **(b)** Hard (**c**) Unique (**d**) None of these

2. ___________ operating system process the data stored on multiple locations.
(a) Parallel **(b)** Network **(c)** Distributed **(d)** Batch

3. __________ provides an interface to the services made available by an operating system.
(a) System Call **(b)** Interrupt (**c**) Process Trace (**d**) None of these

4. Circular wait condition is prevented by defining a __________ordering of resource types.
(a) Normal **(b)** Non Linear (**c**) Random (**d**) Linear

5. _____________is a section of code within a process that requires access to shared resources.
(a) Dispatcher **(b)** Deadlock
(c) Mutual Exclusion **(d)** None of these

6. __________transition of seven state process model shows poor design of the model.
(a) Blocked→Blocked/Suspend **(b)** Blocked/Suspend→Blocked
(**c**) Ready→ Ready/Suspend (**d**) None of these

7. ________number of process states reside in secondary memory in seven state process model.
(a) 3 **(b)** 4 (**c**) 2 (**d**) 5

8. _________fragmentation problem occurs in fixed partitioning.
(a) External **(b)** Locality **(c)** Internal **(d)** None of these

9. A memory management scheme in which secondary memory can act as a part of main memory is called ________.
(a) fixed partitioning **(b)** segmentation
(c) virtual memory **(d)** dynamic partitioning

10. The selection function of HRRN policy is _________.
(a) max(w+s) **(b)** max((w-s)/s)) **(c)** max((w+s)/s) **(d)**max((w*s)/s)

11. _________is a preemptive scheduling policy.
(a) Feedback **(b)** Round Robin **(c)** SRT **(d)** All above

Q.2 [A] Which data structure is used by OS to manage processes? Explain that data structure with all common fields. **[05]**

[B] What do you mean by Semaphore? Give operations of it and give the solution of mutual exclusion using it. **[04]**

[C] Explain the role of OS in program execution and I/O access related services. **[03]**

OR

Q.2[A] Give the answers of following questions based on the Round Robin (RR)(with q=1) and Feedback (with $q=2^i$) scheduling algorithms: **[05]**

(1) What is the finish time of D?
(2) What is the waiting time of C?
(3) What is the normalized turn around time of E?
(4) What is the turnaround time of B?
(5) What is the finish time of F?

The data related to processes are as under :

Process	**Arrival Time**	**Service Time**
A	0	2
B	1	6
C	3	1
D	6	7
E	7	2
F	8	1

[B] What do you mean by deadlock? Explain conditions of it. **[04]**

[C] Explain Physical and Logical Organizations of memory management. **[03]**

Q.3 **Answer the following questions in details**(**Any Four**). **[12]**

1.What do you mean by OS? Give objectives of it.
2.Give solution of the producer-consumer problem using Semaphore.
3.Determine the number of page faults according to Optimal and LRU replacement policies of virtual memory.
The sequence of pages for the execution of process :
1,3,2,4,3,1,2,4,2,1,3,2,1,2,4,3,1,3,2,4,1
The resident set of the process is 3.
4.Give solution of the Dining Philosopher problem that does not lead to the deadlock.
5.Explain paging memory management technique using appropriate example.
6. Draw state transition diagram of five state process model and explain all states of it.

SECTION-2

Q.4 **Fill in the blanks with appropriate answer.** **[11]**

1. **_____________**attribute of file controls who can do reading, writing, and executing.
2. **___________**type of a file having compiled,machine language, not linked functionalities.

3.__________layer of file system manages the metadata information of file.

4.________file allocation is suitable for the direct access kind of file.

5.Printer is an example of a_________readable I/O device.

6. __________ is a collection of electronics that can operate a bus, a port or a device.

7. A cache is region of ________ memory that holds copies of data.

8. Principle of ______________indicates that the user,process and procedure be given just enough privileges to perform their task.

9. _________command is used in Linux to rename the file.

10._______ Linux command display user who uses the Linux.

11. ________Linux command compares two files.

Q.5 [A]	Explain different layers of file system.	**[04]**
[B]	Explain DMA I/O communication technique.	**[04]**
[C]	Write a shell script to find prime numbers between 1 to N.	**[04]**
	OR	
Q.5[A]	Explain any four kernel I/O subsystems.	**[04]**
[B]	Explain contiguous file allocation with example. Discuss which problem will occur in this kind of allocation and how it is to be resolved.	**[04]**
[C]	Write a shell script to accept any N numbers and compute the sum of only even numbers. Use appropriate validations for the same.	**[04]**

Q.6 **Answer following questions in detail** (**Any Four**). **[12]**

1. Explain the steps of file system mounting.

2. Give the characteristics of RAID architecture and explain any one layer having a true RAID characteristics.

3. Explain **ls** command with at least six switches.

4. Explain security violations and list out different levels of security measure.

5. Explain programmed I/O and Interrupt driven I/O techniques.

6. Explain the following Linux commands with appropriate syntax and example:

(a)chmod **(b)**grep **(c)**cat
